SYSC 5108 Exam Study: Assignment 2, Question 3

Logistic regression predictions with one-hot categorical inputs

Source question: `SYSC 5108 W26 Assignment 2.pdf`, Question 3

Question Summary

We are given a trained multivariate logistic regression model that predicts whether a shopper will perform a repeat purchase after receiving a free gift. The input features are:

$$\text{age}, \quad \text{socioeconomic band}, \quad \text{shop value}, \quad \text{shop frequency}.$$

The socioeconomic band has three possible values: a, b, and c. The assignment says to use one-hot coding for b and c, with a as the baseline. Therefore:

$$a \to (B = 0, C = 0), \qquad b \to (B = 1, C = 0), \qquad c \to (B = 0, C = 1).$$

The model weights are:

Feature	Weight
Intercept	−3.82398
Age	−0.02990
Band B	−0.09089
Band C	−0.19558
Shop value	0.02999
Shop frequency	0.74572

The query instances are:

ID	Age	Band	Shop frequency	Shop value
1	56	b	1.60	109.32
2	21	c	4.92	11.28
3	48	b	1.21	161.19
4	37	c	0.72	170.65
5	32	a	1.08	165.39

Step 1: Write the Logistic Regression Model

Chapter 3 slides 32–34 define the logistic sigmoid and logistic regression prediction:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \qquad p(Y = 1 \mid x) = \sigma(z).$$

Goodfellow, Bengio, and Courville also describe logistic regression as using the logistic sigmoid to squash a linear function into the interval $(0, 1)$ so it can be interpreted as a probability.

For this assignment, the linear score is:

$$\begin{aligned} z = {} & w_0 + w_{\text{age}}(\text{Age}) + w_B B + w_C C \\ & + w_{\text{value}}(\text{Shop value}) + w_{\text{freq}}(\text{Shop frequency}). \end{aligned}$$

Substitute the weights:

$$\begin{aligned} z = & -3.82398 - 0.02990(\text{Age}) - 0.09089B - 0.19558C \\ & + 0.02999(\text{Shop value}) + 0.74572(\text{Shop frequency}). \end{aligned}$$

Then compute:

$$p = \sigma(z) = \frac{1}{1+e^{-z}}.$$

If a hard classification is needed, use the usual threshold:

$$p \geq 0.5 \Rightarrow \text{predict repeat purchase}, \qquad p < 0.5 \Rightarrow \text{predict no repeat purchase}.$$

Step 2: Worked Example for ID 1

For ID 1:

$$\text{Age} = 56, \quad \text{Band} = b, \quad \text{Shop frequency} = 1.60, \quad \text{Shop value} = 109.32.$$

Since the band is b:

$$B = 1, \qquad C = 0.$$

Now compute:

$$\begin{aligned} z_1 = & -3.82398 - 0.02990(56) - 0.09089(1) - 0.19558(0) \\ & + 0.02999(109.32) + 0.74572(1.60). \end{aligned}$$

Term by term:

$$\begin{aligned} -0.02990(56) &= -1.67440, \\ -0.09089(1) &= -0.09089, \\ 0.02999(109.32) &= 3.2785068, \\ 0.74572(1.60) &= 1.193152. \end{aligned}$$

Therefore

$$z_1 = -3.82398 - 1.67440 - 0.09089 + 3.2785068 + 1.193152 = -1.1176112.$$

Apply sigmoid:

$$p_1 = \sigma(-1.1176112) = \frac{1}{1+e^{1.1176112}} = 0.2464547.$$

Since $0.2465 < 0.5$, ID 1 is predicted not to repeat purchase.

Step 3: Compute All Predictions

The same calculation is applied to each query instance:

ID	B	C	Age	Freq	Value	z	$\sigma(z)$
1	1	0	56	1.60	109.32	−1.1176	0.2465
2	0	1	21	4.92	11.28	−0.6402	0.3452
3	1	0	48	1.21	161.19	0.3863	0.5954
4	0	1	37	0.72	170.65	0.5289	0.6292
5	0	0	32	1.08	165.39	0.9846	0.7280

Step 4: Make the Gift Decision

Using threshold 0.5:

ID	$\sigma(z)$	Decision
1	0.2465	Do not give gift
2	0.3452	Do not give gift
3	0.5954	Give gift
4	0.6292	Give gift
5	0.7280	Give gift

The highest predicted propensity is ID 5, followed by ID 4 and ID 3. IDs 1 and 2 have probabilities below 0.5 and would not receive the gift under this threshold rule.

Step 5: Exam Interpretation

This question tests three things:

- Logistic regression uses a linear score $z = w^T x$ followed by the sigmoid $\sigma(z)$ to produce a probability.
- One-hot encoding needs a baseline category. Here, category a is represented by $B = 0, C = 0$, not by a separate third indicator.
- The sign and magnitude of each weight affect the log-odds. For example, age has a negative coefficient, while shop value and shop frequency have positive coefficients.

A concise exam answer could be:

> Encode the socioeconomic band as $a = (0, 0)$, $b = (1, 0)$, and $c = (0, 1)$. For each query, compute
>
> $$\begin{aligned} z = & -3.82398 - 0.02990(\text{Age}) - 0.09089B - 0.19558C \\ & + 0.02999(\text{Value}) + 0.74572(\text{Frequency}), \end{aligned}$$
>
> then compute $p = \sigma(z) = 1/(1 + e^{-z})$. The resulting probabilities for IDs 1 through 5 are 0.2465, 0.3452, 0.5954, 0.6292, and 0.7280. With a 0.5 threshold, give the free gift to IDs 3, 4, and 5, but not to IDs 1 and 2.

Cross References Used

- Assignment: `SYSC 5108 W26 Assignment 2.pdf`, Question 3.
- Local submitted Assignment 2 PDF checked for comparison, Question 3 response.
- Local spreadsheet/notebook checked: `A2_Q3.ods` and `A2_Q3.ipynb`.
- Slides: Chapter 3 slide deck, slides 32–34 on the logistic sigmoid, binary logistic regression, logit interpretation, and prediction.

- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3, Section 3.10 on the logistic sigmoid.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 5, logistic regression discussion around Equation 5.81.